

Tasfia Akter Sara

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217 N 35th St Philadelphia, PA 19104, USA

EDUCATION

- **Military Institute of Science and Technology, Bangladesh University of Professionals** Dhaka, Bangladesh
B.Sc. IN COMPUTER SCIENCE AND ENGINEERING Jan 2020 - Apr 2024
 - CGPA: **3.83/4.00** (ranked 4th out of 88 students)
 - Undergraduate Thesis: An Ensemble-Based Deep Neural Approach for Classifying Skin Allergies
 - IELTS Score: **7.5** (L: 8.5, R: 8.5, W: 7, S: 6)

RESEARCH INTEREST

Human Computer Interaction, Machine Learning, Artificial Intelligence, Digital Image Processing, Data Science, IoT, Computer Vision

PROFESSIONAL EXPERIENCE

- **Former Lecturer, Department of Computer Science and Engineering** Dhaka, Bangladesh
Military Institute of Science and Technology Apr 2024 – Aug 2026
- **Industrial Trainee** Dhaka, Bangladesh
Brain Station 23 2023
- **Management Lead, MIST Mars Rover Society** Dhaka, Bangladesh
Military Institute of Science and Technology 2022 - 2023

PROJECTS

- **Hydroferma (Capstone Project)** 2022 - 2023
Tools and Technologies: Machine Learning, Image Processing, Firebase, Flutter, Python, Arduino, Data Analysis, IoT sensors
 - An automated ML-based hydroponic vertical farm monitoring system (with mobile application)
- **Unlocking Voices** 2023 - 2024
Technologies: HCI, HE, AE, WCAG
 - Evaluation of the usability & accessibility of mobile applications for stutter reduction and communication enhancement in children
- **PPDHero** 2023 - 2024
Technologies: Information Analysis
 - A research to empower new mothers & raise awareness about postpartum depression
- **Unraveling the Threads** Current
Technologies: Machine Learning, Image Processing, Python, Data Analysis
 - A comparative study of jamdani and non-jamdani fabric classification using deep learning
- **The Last of Us** 2023
Tools: Blender
 - A graphic-based post-apocalyptic world simulation
- **Conveyor Belt Infrared Counter** 2022
Technologies: Arduino and IoT Sensors
 - Conveyor belt infrared counter with Arduino integration for object counting and display using motors, IR sensor, and LED display.
- **MCC Virtual Assistant** 2022
Tools and Technologies: HTML, CSS, PHP
 - A web-based management system for MIST Computer Club (MCC).
- **Subscriptia- An E-commerce Application, Student Study Portal** 2022
Technology: Python, Django Framework, CSS, Firebase

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java, PL/SQL, Assembly (Intel x86)
- **Tools:** Blender, OpenGL, Google Colab, Visual Studio, Codeblocks, Matlab, Android Studio, Git
- **Database:** Oracle, FireBase
- **Simulation Softwares:** Cisco Packet Tracer, Circuit Maker
- **OS:** Linux, Windows
- **Embedded Systems:** Arduino, ESP32
- **Other Tools & Technologies:** Latex, MS Word, MS Project, MS PowerPoint, MS Excel, Canva

PUBLICATIONS

C=Conference, J=Journal on process, S=In Submission, A=Accepted

- [C.1] T. A. Sara, S. N. Nazifa, S. Tasneem, T. U. Shakib, & M. N. Islam, (2023, December). "PPDHero: Requirements Elicitation and Development of a System to Empower New Mothers on Postpartum Depression." In International Conference on Intelligent Systems Design and Applications (pp. 389-398). Cham: Springer Nature Switzerland
- [C.2] S. N. Nazifa, S. T. Mulk, T. A. Sara, T. U. Shakib and M. N. Islam, "Unlocking Voices: Assessing the Usability and Accessibility of Mobile Applications for Stutter Reduction," 2024 27th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2024, pp. 1016-1021.
- [C.3] T. A. Sara, B. Ghosh, S. N. Nazifa and H. Jahan, "An Ensemble-Based Deep Neural Approach for Classifying Skin Allergies," 2025 28th International Conference on Computer and Information Technology (ICCIT), 2025, pp. 3500-3505.
- [A.1] T. Mahamud, S. W. Tawassaf, Sara, T. A, H. Rahman, and MD. I. Salekin, "Machine Learning Model for Predicting Mechanical Behavior of I-Beams with Varying Dimensions." 3rd ICMMPPE 2025, DUET, Dhaka. (In press)
- [J.1] T. A. Sara, K. H. Irfan, M.R.U. Islam and H. Jahan "A Hybrid DINO-ViT and Quantum Annealing Framework for Skin Lesion Classification"

HONORS AND AWARDS

- **Commandant's List of Honour** 2020 - 2024
Military Institute of Science and Technology
 - Achieved CGPA 3.80 or above in all the four academic years
- **Dean's List of Honour** 2020 - 2024
Military Institute of Science and Technology
 - Achieved CGPA 3.70 or above in all the four academic years
- **Best Technical Paper Presentation** 2024
International Conference on Computer and Information Technology (ICCIT)
 - Got the best technical presentation award of the session for the paper titled "Unlocking Voices: Assessing the Usability and Accessibility of Mobile Applications for Stutter Reduction"
- **Champion in University Rover Challenge 2021** 2021
The Mars Society, USA
 - Team Member of MIST MONGOL BAROTA
- **2nd Runnerup in Anatolian Rover Challenge 2022** 2022
Space Exploration Society(UKET), Turkiye
 - Management Lead of MIST MONGOL BAROTA

TEACHING EXPERIENCE

- **Theory Courses:**
 - CSE - 205 Object Oriented Programming, MIST, Bangladesh, Lecturer(85 students), Spring 2026, Spring 2025
 - CSE - 217 Theory of Computation, MIST, Bangladesh, Lecturer(60 students), Spring 2024
 - CSE - 109 Computer Programming, MIST, Bangladesh, Lecturer(100 students), Fall 2025, Fall 2024
- **Sessional Courses:**
 - CSE - 206 Object Oriented Programming Sessional, MIST, Bangladesh, Lecturer(85 students), Spring 2026, Spring 2025
 - CSE - 204 Data Structure and Algorithm Sessional, MIST, Bangladesh, Lecturer(45 students), Spring 2026
 - CSE - 302 Database Management System, MIST, Bangladesh, Lecturer(45 students), Spring 2024
 - CSE - 308 Operating System, MIST, Bangladesh, Lecturer(45 students), Spring 2024
 - CSE - 106 Structured Programming Language, MIST, Bangladesh, Lecturer(105 students), Fall 2024
 - CSE - 110 Computer Programming, MIST, Bangladesh, Lecturer(100 students), Fall 2025, Fall 2024
 - CSE - 104 Digital Logic Design Sessional (85 students), Fall 2025

ACADEMIC REFERENCES

1. **Dr. Md. Mahbubur Rahman**
Professor, Computer Science and Engineering, Military Institute of Science and Technology
Email: mahbub@cse.mist.ac.bd
2. **Dr. Muhammad Nazrul Islam**
Associate Professor, Computer Science and Engineering, Military Institute of Science and Technology
Email: nazrul@cse.mist.ac.bd